

Build first CRUD API

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Track: FlyRank AI Internship · Backend

Week: 2 Assignment A1

Project

Task API — a FastAPI CRUD API for managing to-do tasks.

GitHub Repository

https://github.com/JIYA-YDV/flyrank-internship-tracker/tree/main/week_2/flyrank-todo-api

Note: This assignment lives inside my consolidated FlyRank internship tracker repo. Stage-by-stage commits are visible in the commit history.

What I Built

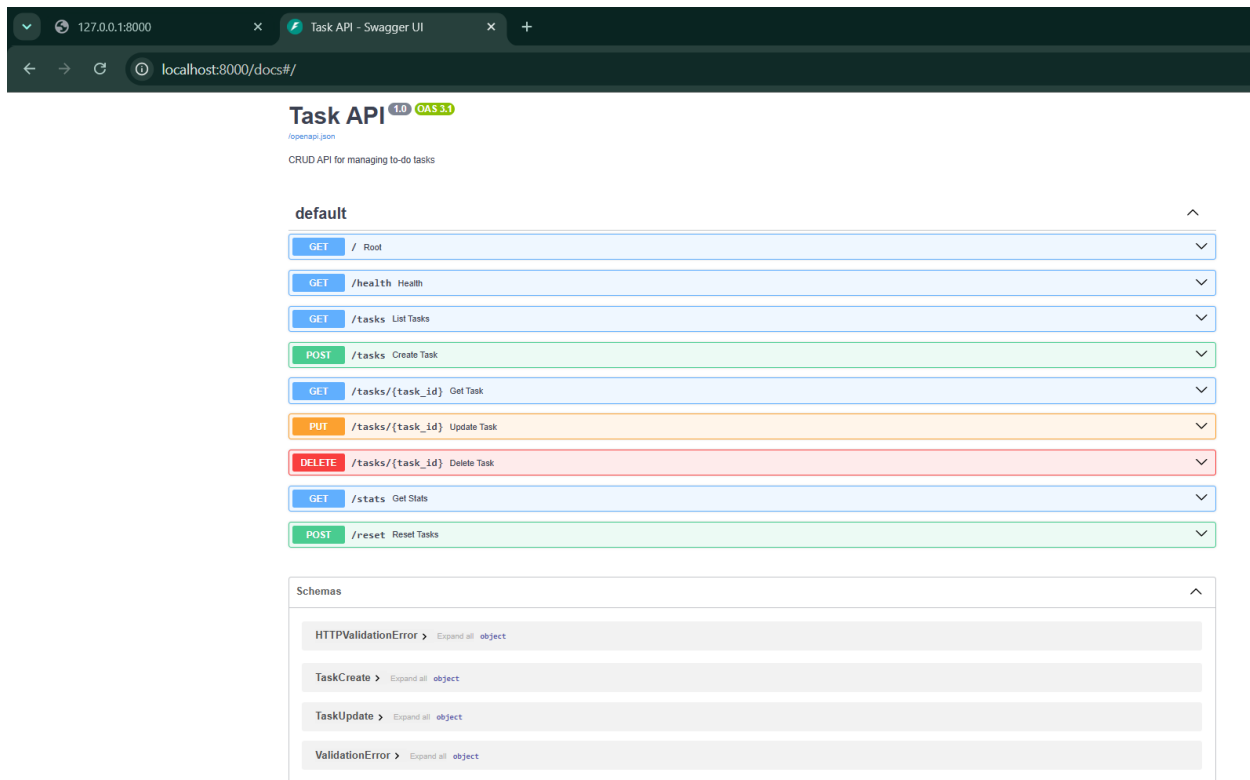
Small REST API in Python + FastAPI. Manages an in-memory to-do list and supports full CRUD.

Endpoints:

- GET / — API information
- GET /health — health check
- GET /tasks — list all tasks (supports ?done= and ?search= filters)
- GET /tasks/{task_id} — get one task by ID
- POST /tasks — create a new task
- PUT /tasks/{task_id} — partial update (title and/or done)
- DELETE /tasks/{task_id} — delete a task
- GET /stats — task statistics (stretch)
- POST /reset — reset seed data (stretch)

Testing Evidence

Verified full CRUD cycle twice: once via curl.exe in PowerShell, once via Swagger UI "Try it out" at <http://localhost:8000/docs>



Final status code checks:

| Request | Result |

|---|---|

| GET /health | 200 OK |

| GET /tasks | 200 OK |

| GET /tasks/1 | 200 OK |

| GET /tasks/99 | 404 Not Found |

| POST /tasks (valid body) | 201 Created |

| POST /tasks (missing title) | 422 Unprocessable Entity |

| PUT /tasks/1 (valid body) | 200 OK |

| PUT /tasks/1 (empty body) | 400 Bad Request |

| DELETE /tasks/4 | 204 No Content |

| GET /tasks/4 after delete | 404 Not Found |

| GET /stats | 200 OK |

Swagger screenshot is embedded in the README.

Bonus Stage 7 Completed

Wrote a prompt from memory, generated an AI version, ran both APIs in parallel on ports 8000 and 8001, and documented three concrete differences in the "AI vs Me" README section. Key finding: the AI made PUT title required, which breaks partial updates — a real bug caused by my prompt not specifying PUT semantics.

PowerShell / curl.exe Note

Used curl.exe rather than PowerShell's curl alias. Because Windows PowerShell can corrupt JSON quoting when passing inline bodies to curl.exe, I wrote JSON bodies to files and used -d "@file.json" for reliable testing:

```
[System.IO.File]::WriteAllText("$PWD\create.json", '{"title":"Final check"}')
```

```
curl.exe -i -X POST http://localhost:8000/tasks `
```

```
-H "Content-Type: application/json" `
```

```
-d "@create.json"
```